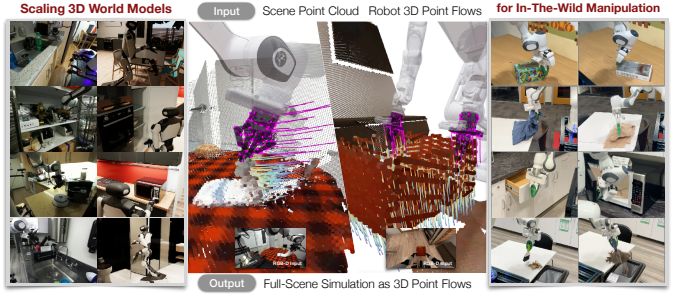
PointWorld: Scaling 3D World Models for In-The-Wild Robotic Manipulation

Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, Li Fei-Fei.
Stanford, NVIDIA. CVPR 2026. arXiv 2601.03782. Code, data and checkpoints: github.com/NVlabs/PointWorld

Juntang Wang · Vivian Hui

Problem



Huang et al. (2026), Fig. 1

- A **world model** for manipulation: given what the robot sees and what it intends to do, predict how the scene will move
- The goal here: from one RGB-D image of an unseen room, in 3D, for more than one robot
- Represent the scene and the robot's action the same way, as moving 3D points, and train one model on 2M trajectories

Before PointWorld

Physics simulators

(Todorov et al., 2012)

- Accurate once the scene is modelled: meshes, masses, materials
- That modelling is per scene, and the sim-to-real gap remains

Learned particle dynamics

(Li et al., 2018; Ai et al., 2025)

- Learn from interaction, often as graphs over particles (GBND)
- Usually scene-specific, with full observability, object or material priors

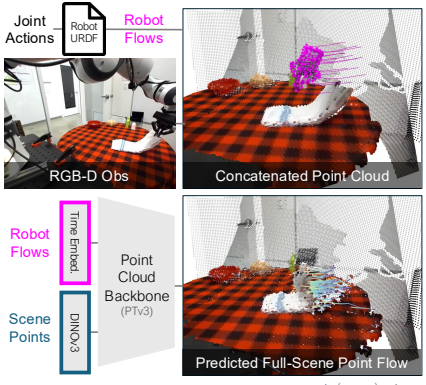
Video world models

(Brooks et al., 2024; Agarwal et al., 2025; Assran et al., 2025)

- Trained at scale and look real, but physics is often off
- Actions come in as text or robot commands, not geometry; seconds per prediction

PointWorld keeps the particle view, and scales it the way video models scale

Method

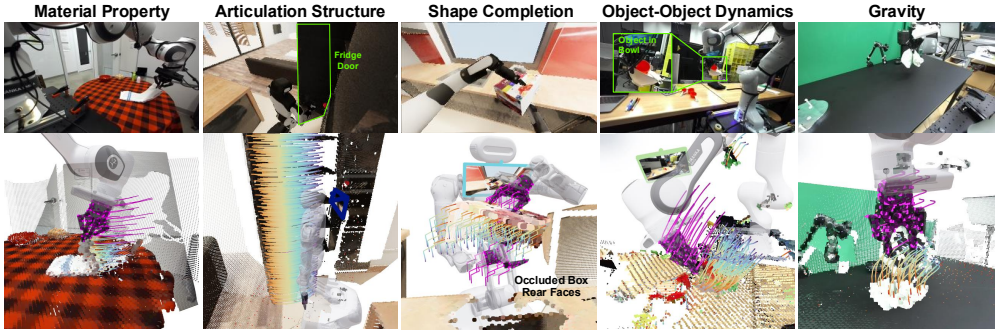


Huang et al. (2026), Fig. 2

- **State**: the RGB-D image back-projected to a point cloud, robot pixels removed. Each point carries frozen DINOv3 features (Siméoni et al., 2025)
- **Action**: 300 to 500 points per gripper, placed by forward kinematics at each of the next 10 steps and tagged with a time embedding. A Franka and a two-armed humanoid are described the same way
- **Model**: PTv3 (Wu et al., 2023) reads scene and robot points as one cloud; a shared head gives every scene point's displacement at all 10 steps, 0.1 s apart, in one pass

State and action live in one space: 3D points

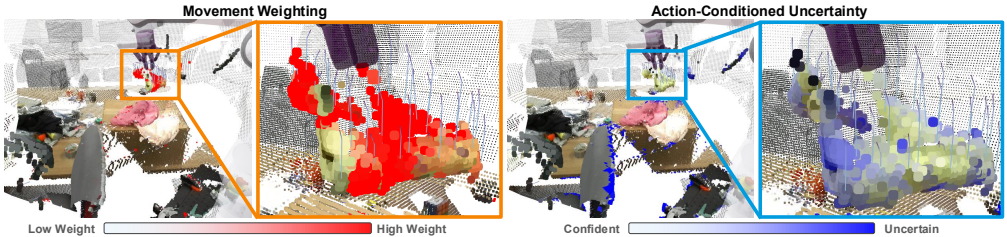
What one target asks for



Huang et al. (2026), Fig. 3

- To predict where every point goes, the model has to work out what is an object, what is soft or hinged, the hidden back of a box, and gravity
- Training gives no object masks, material labels or physics engine: only the point displacements, helped by frozen DINOv3 features

Training loss

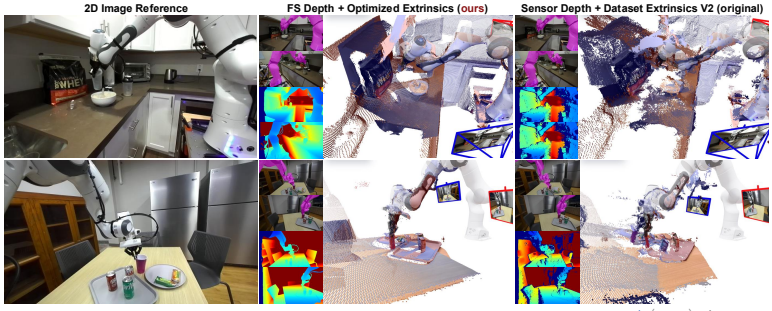


Huang et al. (2026), Fig. 4, bottom row

$$\mathcal{L} = \frac{1}{2} \sum_{k,i} w_{k,i} \left(\rho_\delta \left(\hat{P}_{t+k,i} - P_{t+k,i} \right) e^{-s_{k,i}} + s_{k,i} \right)$$

- Only 1–5% of points move. w shifts the loss onto them, using the true displacement
- Real labels are noisy. The model predicts its own log-variance s per point, which down-weights bad labels, and ρ_δ is a Huber loss
- s also picks up physics: it is largest along the edge of a dropped cloth

Data: re-annotating DROID in 3D



Huang et al. (2026), Fig. 5, top

- DROID (Khazatsky et al., 2024): real teleoperation, but its depth and camera poses are off. Three fixes: depth from FoundationStereo (Wen et al., 2025), poses from VGGT (Wang et al., 2025) refined against the robot mesh, CoTracker3 (Karaev et al., 2025) tracks lifted to 3D
- Over 60% of DROID survives, about 200 hours. With BEHAVIOR-1K simulation (Li et al., 2024), the total is about 2M trajectories, 500 hours

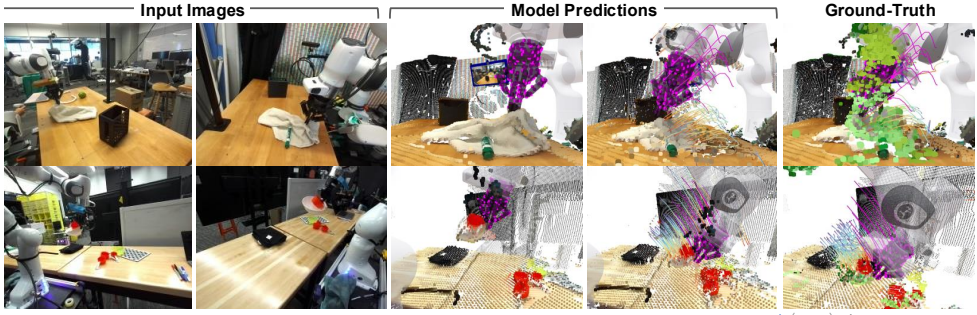
Planning with the model

$$\min_{E_{0:T}} \sum_{k=1}^{T} c_{\text{task}}(s_k) + c_{\text{ctrl}}(E_k) \quad \text{s.t.} \quad s_{1:T} = F_\theta(s_0, a_{1:T})$$

E_k: end-effector pose · s_k: predicted scene points · a_k: gripper points placed at E_k

- MPPI (Williams et al., 2017): perturb a gripper path, roll each one out through PointWorld, average the paths weighted by how low their cost is
- The cost: move the points of a hand-drawn object mask to hand-placed targets, plus a mild pull toward the object, path length and reachability
- One image in, a plan 30 steps (3 s) long out, in a few seconds

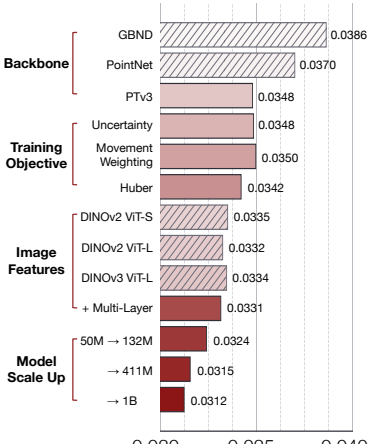
How it is scored



Huang et al. (2026), Fig. 6, top two rows

- ℓ_2 **mover**: mean distance, in metres, between predicted and true positions of the points that move, over the 1 s horizon
- On DROID, points count only where a separate model, trained on the test split, is confident (§4: the top 80%)
- 0.001 is one millimetre. Read every difference that follows in mm

Recipe: what each change bought

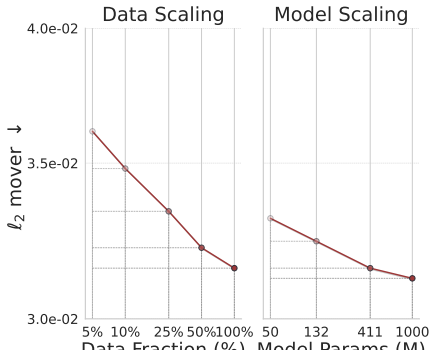


Huang et al. (2026), Fig. 7

- Swapping the graph network (GBND) for PTv3 is the biggest single step: 0.0386 to 0.0348
- The loss and DINO image features add 1.7 mm; scaling PTv3 from 50M to 1B parameters adds 1.9 mm
- 0.0386 to 0.0312 overall: 7.4 mm, or 19% less error on moving points

Each bar adds one change to the bar above it. Hatched bars were tried and not kept.

Scaling

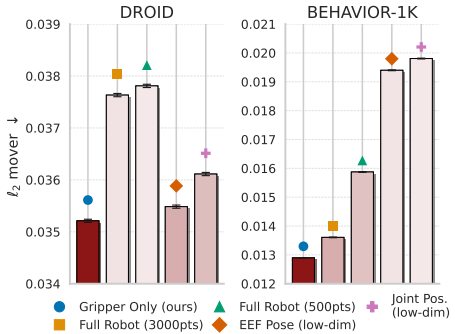


Huang et al. (2026), Fig. 9

- Error falls roughly log-linearly with data, from 5% to 100% of DROID, and with model size, from 50M to 1B parameters
- PTv3 makes the model axis affordable: 957 times GBND's parameters for 4.3 times the memory, about 0.12 s per pass at 1B (Tab. 1)
- Predictable gains from more data and capacity, as in language and vision

One axis at a time, on DROID. All nine backbones in the backup.

Why gripper points?



Huang et al. (2026), Fig. 11

- Robot as points on its grippers only, against points on the whole body (two densities), end-effector pose, or joint positions
- Lowest error on both real (DROID) and simulated (B1K) data. On real data, whole-body points do worse than a plain pose; the authors' reading: too many robot points drown the few moving scene points
- Gripper points keep the contact geometry and stay the same format for both robots

All variants are trained on DROID and B1K together. The paper does not say what the markers above the bars are.

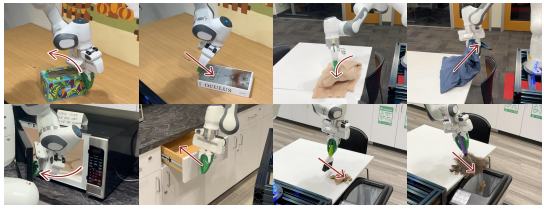
Does it transfer to a new lab?

pretrained on	DROID	BEHAVIOR-1K	both	specialist (H only)
zero-shot	0.0305	0.0531	0.0300	0.0293
finetuned	**0.0271**	0.0299	**0.0272**	0.0293

Huang et al. (2026), Tab. 2, held-out columns, ℓ_2 mover; in- and cross-domain columns in the backup

- H: one DROID lab held out of pretraining. The specialist trains on that lab alone
- Zero-shot, real-plus-sim pretraining comes within 0.7 mm of the specialist. Finetuned with 1/20 of the training steps, it beats the specialist by 2.1 mm
- Simulation alone does not transfer zero-shot, but finetuning brings it close to the specialist

Results: real robot, zero-shot



Huang et al. (2026), Fig. 8

- 8 tasks × 10 trials, one camera, a gripper never seen in training, no demos or finetuning: 56% overall (45 of 80)
- Each plan runs without replanning; the deformable and tool tasks start with the object grasped. Videos: point-world.github.io

Limitations

Stated by the authors (Appendix)

- One still frame in, so the scene must start at rest: no velocities
- Task costs are drawn by hand: object masks and target points
- The robot is assumed rigid and to follow its planned path exactly

Takeaway

- One 3D world model for more than one robot: scene and action as moving points, trained on 2M real and simulated trajectories
- Each piece of the recipe lowers the error, and more data and parameters keep helping at a predictable rate
- From a single image it plans real pushing, cloth, hinge and tool tasks zero-shot; code, data and checkpoints are public

Discussion

- The robot results report no baseline and never replan. What comparison would show that 56% is good?
- On real data a plain end-effector pose comes close to gripper points. Is the point action essential, or mainly a way to mix robots?
- The model sees one frozen frame, and moving points are still off by about 3 cm. Which tasks does that rule out?

Backup: our reading

- **Robot**: 8 tasks $\times$ 10 trials, no baseline success rate, no replanning, deformable and tool tasks start pre-grasped. 56% cannot yet be ranked against a policy or another model
- **Metric**: real-data gaps are a few mm, scored on points an expert model trusts. §4 says it keeps the top 80%; the appendix thresholds "at the 0.8 quantile" and drops the points below, which reads as the top 20%
- **Transfer**: zero-shot on a held-out lab is 0.7 mm behind the specialist; the clear wins need finetuning
- **Small inconsistencies**: GBND is 0.0390 in Tab. 1 and 0.0386 in Fig. 7. Fig. 11's legend gives whole-body variants of 500 and 3000 points; §5.2 gives "the same number of points" and 2000

Backup: training objective

$$\frac{1}{2}\sum_{k,i}^{H,N_S} \underbrace{w_{k,i}}_{\text{movement}} \Big(\underbrace{\rho_\delta\big(\hat{P}_{t+k,i} - P_{t+k,i}\big)}_{\text{Huber on 3D residual}} \underbrace{e^{-s_{k,i}}}_{\text{uncertainty weight}} + \underbrace{s_{k,i}}_{\text{regularizer}} \Big) \tag{1}$$

$$m_{k,i} = \sigma\big(\kappa(\delta_{k,i} - \tau)\big), \qquad w_{k,i} = m_{k,i} \Big/ \textstyle\sum_{k,i} m_{k,i}$$

Huang et al. (2026), §3.1

- $\delta_{k,i}$: length of point i's true displacement at step k; τ a threshold, κ a temperature
- Points the 2D tracker marks as occluded get no loss at all
- Aleatoric uncertainty, the same device VGGT uses (Wang et al., 2025): the model may say "unsure", at the price of the s term

Backup: planning and robot protocol

Planner

- MPPI around a zero nominal, cubic-spline noise (4 knots), σ from 0.05 to 0.50
- 256 samples, temperature $\beta = 0.05$, EMA 0.9, 20 iterations
- 30 steps ahead, in 3 chunked passes; no replanning in this work

Robot

- 7-DoF Franka on a wheeled base, 3D-printed fin-ray gripper (training saw Robotiq and Galaxea R1 Pro grippers)
- One RealSense D435; depth from its stereo IR pair via FoundationStereo
- Masks from SAM 2 (Ravi et al., 2024), targets placed in a GUI; failures include plans judged unsafe to run

Huang et al. (2026), Appendix, real-robot experiment details

Backup: all backbones

	params	memory	FLOPs	latency (ms)	ℓ_2 mover	ℓ_2 static
GBND (Ai et al., 2025)	1.00×	1.00×	1.00×	13.46	0.0390	0.0066
PointNet	1.03×	0.34×	0.04×	5.93	0.0369	0.0084
PointNet++	1.07×	0.67×	0.06×	327.08	0.0368	0.0073
SparseConv	33.31×	7.18×	1.32×	17.70	0.0396	0.0076
Transformer	41.06×	0.31×	3.38×	30.43	0.0339	0.0071
PTv3-50M	49.14×	0.30×	0.34×	59.60	0.0331	0.0067
PTv3-132M	127.22×	0.69×	1.04×	69.60	0.0324	0.0061
PTv3-411M	398.67×	1.89×	1.90×	102.47	0.0315	0.0059
PTv3-1B	957.71×	4.30×	3.57×	123.65	**0.0312**	**0.0056**

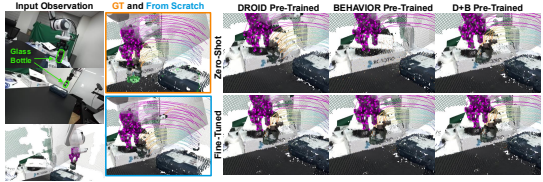
Huang et al. (2026), Tab. 1, relative to GBND

- GBND keeps a feature for every point and passes messages only locally, so memory grows fast and far-away effects get lost

Backup: generalization, all columns

		D→D	B→B	D→B	B→D	D→H	B→H	D+B→H	specialist
zero-shot	mover	0.0315	0.0087	0.1460	0.0558	0.0305	0.0531	0.0300	0.0293
	static	0.0059	0.0010	0.0050	0.0058	0.0049	0.0057	0.0063	0.0043
finetuned	mover	–	–	0.0107	0.0378	0.0271	0.0299	0.0272	0.0293
	static	–	–	0.0003	0.0086	0.0040	0.0046	0.0040	0.0043

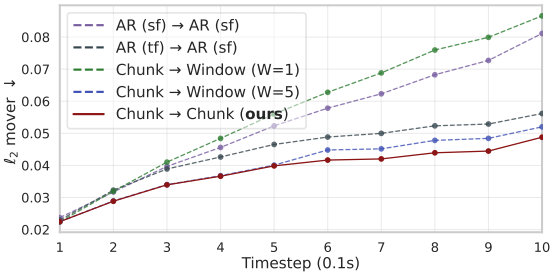
Huang et al. (2026), Tab. 2: D = DROID, B = BEHAVIOR-1K, H = held-out lab



Huang et al. (2026), Fig. 10

H is the CLVR lab within DROID, split 90/10; finetuned models see only the 90%. Zero-shot across sim and real fails both ways (0.1460, 0.0558).

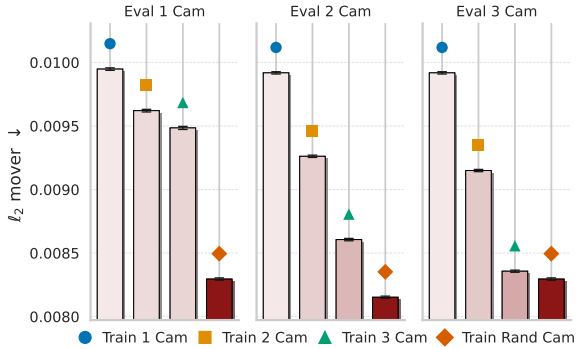
Backup: predict 1 s at once



Huang et al. (2026), Fig. 12

- Predicting all 10 steps together drifts least, in one pass instead of 2 to 10
- The same model fed its own outputs one step at a time drifts most
- The planner chains three such chunks to look 3 s ahead

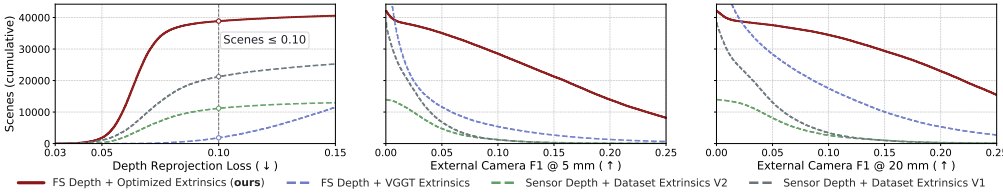
Backup: cameras



Huang et al. (2026), Fig. 13

- More cameras in training help at every test setting
- Training on a random draw of up to three cameras is the most robust

Backup: annotation quality



Huang et al. (2026), Fig. 5, bottom

- Left: depth reprojection error on the robot's own surface. Right: point cloud agreement between two cameras within 5 and 20 mm
- Off-the-shelf models alone (FoundationStereo, VGGT) are not enough; the robot-mesh refinement is what lifts the curves
- Extrinsics land a median 1.8 cm and 1.9° from a proxy truth: the best 1% of scenes under DROID's own calibration

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